NANYANG TECHNOLOGICAL UNIVERSITY

School of Social Sciences

HE3002 Macroeconomics III

LECTURE 1 pp1-6

The Open Economy: Openness in Goods and Financial Markets

READINGS

1. Olivier Blanchard *Macroeconomics,* 8th Edition (2021), Prentice Hall, Chapter 17.

INTRODUCTION

1. HE2002 Macroeconomics II covers topics in which the economy we looked at is closed i.e. it does not interact with the rest of the world.

2. However, given the degree of integration in the world economy, a crisis that started in one country, can affect nearly every country in the world. For example, the 2008 global financial crisis.

   [Figure 1: Growth in advanced and emerging economies since 2000]

3. Therefore, it is simply not realistic to assume closed economy.

   Note: $Y = C + I + G + (X - M)$

4. We now turn to openness which has three distinct dimensions:

   (a) Openness in goods markets - the ability of consumers and firms to choose between domestic goods and foreign goods.

   C M i

   (b) Openness in financial markets - the ability of investors to choose between domestic assets and foreign assets. i^*

   labor/workers

   (c) Openness in factor markets - the ability of workers to choose where to work and of firms to choose where to locate production.

   i & i^*

5. The first choice is governed by the relative price of foreign goods; the second by relative returns on foreign assets.

6. In the short run and in the medium run, openness in factor markets plays much less of role than openness in either goods markets or financial markets, therefore, we shall ignore openness in factor markets.

7. For most countries, open economy considerations have long had substantial effects on economic performance and open economy issues are becoming increasingly important.

TB = X − M

8. Lecture 1 focuses on the basic determinants of the trade balance and describes the implications of arbitrage between domestic and foreign bonds. Y = C+I+G closed

9. Lecture 2 integrates the trade balance discussion into the closed economy goods market model (the Keynesian cross). Y = C+I+G+(X−M) → IS goods mkt

money

10. Lecture 3 integrates the asset market discussion into the closed economy model of the money market, and develops an open economy IS-LM model. IS-LM

Goods ↗ ↖ Money market

11. After that, in Lecture 4, we will consider a medium run-model of an open economy with a fixed exchange rate, explore exchange rate crises and the behavior of flexible exchange rates, and discuss the choice of exchange rate regime. Singapore – Managed float

HK – fixed exchange rate regime

Japan – flexible exchange rate regime

X M

OPENNESS IN GOODS MARKETS Exports and Imports

M X

1. An open economy buys from and sells to the rest of the world. This means that consumers in the open economy are required to choose between domestic goods and foreign goods.

2. As a result, in such decision making, it is important to look at the role of the relative price of domestic goods in terms of foreign goods i.e. the real exchange rate. ε E ← Nominal exchange rate

3. It should be noted that even countries most committed to free trade have **tariffs** (taxes on imported goods) and **quotas** (restrictions on the quantity of goods that can be imported).

4. Figure 2 shows the evolution of US exports and imports as ratios of GDP from 1960 to 2017.

[Figure 2: US exports and imports as ratios to GDP since 1960]

5. Trade balance, TB, is the difference between exports, X, and imports, M i.e. $TB = X - M$;

   (a) If exports exceed imports $X > M$, there is a **trade surplus** or equivalently, a **positive trade balance**, $TB > 0$

   (b) If exports is less than imports $X < M$, there is a **trade deficit** or equivalently, a **negative trade balance** $TB < 0$, and

   (c) If exports equals imports $X = M$, there is a **balance trade**, $TB = 0$.

6. Trade openness is usually measured as:

Degree of trade

$$\text{Openness} = \frac{X + M}{Y} \tag{1}$$

where Y is the total GDP of a country.

7. Another measure of openness is the proportion of aggregate output composed of tradable goods, goods that compete with foreign goods in either domestic markets or foreign markets.

8. Examples of **tradable goods** are cars, computers, pharmaceuticals, etc. Examples of **nontradable goods** are housing, medical services, haircuts, etc.

9. Table 1 shows the ratios of exports to GDP for selected OECD countries in 2017.

[Table 1: Ratios of exports to GDP for selected OECD Countries, 2017]

10. Can exports exceed GDP? Export ratio $= \frac{X}{Y} = \frac{X}{GDP} > 1 \Rightarrow X > GDP$

    (a) Since a country cannot export more than it produces, will the export ratio always be less than one?

    $X > GDP$

    (b) No, exports may be larger than GDP because exports and imports may include exports and imports of intermediate goods.

    (c) In 2017, the ratio of exports to GDP in Singapore was 173.

Foreign Exchange Market and Exchange Rates

1. **Foreign exchange market** is a market that involves buying and selling foreign currency.

   $P_{S\$}$ in terms of US\$ $\quad E = \frac{US\$}{S\$} = P_{S\$}$

2. **Nominal exchange rate** is the price of the domestic currency in terms of foreign currency.

   $E\uparrow \rightarrow$ S\$ app^n

   flexible

   (a) **Nominal appreciation** is an increase in the price of the domestic currency in terms of a foreign currency, i.e., an increase in the exchange rate.

   (b) **Nominal depreciation** is a decrease in the price of the domestic currency in terms of a foreign currency, i.e., a decrease in the exchange rate.

3. Fixed exchange rate system is a system in which two or more countries maintain a constant exchange rate between their currencies.

   fixed

   (a) **Revaluations** are increases in the exchange rate, and

   (b) **Devaluations** are decreases in the exchange rate.

[Figure 3: The nominal exchange rate between USD and British Pound]

ε

4. **Real exchange rate** is the price of domestic goods relative to foreign goods.

   (a) **Real appreciation** is an increase in the real exchange rate, i.e., an increase in the relative price of domestic goods in terms of foreign goods.

Real dep^n → P of domestic goods ↓
→ P of foreign goods ↑

(b) **Real depreciation** is a decrease in the real exchange rate, i.e., a decrease in the relative price of domestic goods in terms of foreign goods.

5. The real exchange rate, the price of U.S. goods in terms of British goods, is

foreign

$E = \frac{£}{US\$}$ domestic ← US$

$$\epsilon = \frac{EP}{P^*} \quad (2)$$

$\varepsilon = \frac{\text{Price of domestic goods}}{\text{price of foreign goods}}$

$EP = \frac{£}{US\$} \cdot US\$$

where ϵ is the real exchange rate, E is the nominal exchange rate, P and P^* are domestic price and foreign price, respectively.

$\varepsilon = \frac{EP}{P^*}$

[Figure 4: The construction of real exchange rate]

① E↑ domestic currency app (E↑→ε↑)

ε app^n → Domestic goods are relatively more expensive

[Figure 5: Real and nominal exchange rates between the US and the UK]

② P↑ (P↑→ε↑)

③ P*↓ (P*↓→ε↑)

6. Bilateral exchange rates are exchange rates between two countries.

7. Multilateral exchange rates say the multilateral real U.S. exchange rate (or U.S. real exchange rate) requires data on the geographic composition of U.S. trade for both exports and imports.

[Figure 6: The U.S. multilateral real exchange rate since 1973]

OPENNESS IN FINANCIAL MARKETS (UIP)

1. Openness in financial markets allows financial investors to hold both domestic and foreign assets. bonds

2. A country's openness in financial markets has an important implication. It allows the country to run trade surpluses and trade deficits.

The Choice between Domestic and Foreign Assets

1. Openness in financial markets implies that people face a new financial decision: whether to hold domestic assets or foreign assets.

2. Consider the choice between US (i) and UK (i^*) one-year bonds from the point of view of a US investor:

   (a) Suppose that you have \$1,000 and decide to hold US one-year bonds.

   (b) The interest rate in the US i_t is 2%. Then, for \$1,000, you buy US bonds, you will get \$1,020 which is obtained by \$1,000×(1+2%) next year.

   (c) Suppose that you decide to hold UK bonds instead.

   (d) To buy UK bonds, you must first buy pounds. Let the exchange rate, E_t, be \$1 for £0.75. For \$1,000, you will get £750.

3. Let interest rate in UK, i_t^*, be 5%. When next year comes, you will get £750×(1+5%)=£787.5. 1.05

4. You will then have to convert your pounds back into dollars. E_{t+1}^e

5. If you expect nominal exchange rate next year, E_{t+1}^e, to be \$1 for £0.78, then each pound will be worth \$(1/0.78)=\$1.28. \$1 ≡ £0.78

6. So, you can expect to have £750×(1+5%)=£787.5 which is £787.5×\$1.28=\$1,010. \$1.28 ≡ £1

7. Now let establish the relationship between interest rates and exchange rates formally.

[Figure 6: Uncovered interest parity (UIP)]

UK \$1010
US \$1020

   (a) Suppose you decide to hold US bonds.

   (b) Let i_t be the one-year US nominal interest rate. Then for every \$1 you put in US bonds, you will get $\$(1 + i_t)$ next year.

   (c) Suppose you decide to hold UK bonds instead.

   (d) To buy UK bonds, you must first buy pounds. Let E_t be the nominal exchange rate between the dollar and pound. For every dollar, you get E_t pounds.

8. Let i_t^* denote the one-year nominal interest rate on UK bonds (in pounds). When next year comes, you will get $E_t(1 + i_t^*)$ pounds.

9. You will then have to convert your pounds back into dollars.

10. If you expect the exchange rate next year to be E_{t+1}^e, then each pound will be worth $(1/E_{t+1}^e)$ dollars.

11. So, you can expect to have $E_t(1 + i_t^*)/E_{t+1}^e$ dollars next year from every dollar you invest now.

12. Assume that you only care about the expected rate of return and hence want to hold only the asset with the highest expected return.

13. In this case, if both UK and US bonds are to be held, they must have the same expected rate of return. This means that:

$$\underbrace{(1 + i_t)} = (E_t)\overbrace{(1 + i_t^*)}^{£}(\frac{1}{E_{t+1}^e}) \quad \text{UIP} \tag{3}$$

$$(1 + i_t) = (1 + i_t^*)(\frac{E_t}{E_{t+1}^e}) \tag{4}$$

14. This is called the **uncovered interest parity condition** or simply the **interest parity condition**.

15. The assumption that financial investors will hold only the bonds with the highest expected return is obviously too strong, for two reasons:

(a) Transaction costs are not considered.

(b) Risk is also not taken into consideration.

$$\frac{E_t}{E_{t+1}^e} = \frac{1}{1+\frac{E_{t+1}^e - E_t}{E_t}}$$

Interest Rates and Exchange Rates

1. Rewrite E_t/E_{t+1}^e as $1/[1+(E_{t+1}^e - E_t)/E_t]$ and replacing in Equation (7) gives:

$$(1+i_t) = \frac{(1+i_t^*)}{1/[1+(E_{t+1}^e - E_t)/E_t]} \tag{5}$$

$(1+x)(1+y) = 1+x+y+\underset{0}{xy}$

$\frac{(1+x)}{(1+y)} =$

2. This gives a relation between domestic nominal interest rate, i_t, the foreign interest rate, i_t^*, and the expected rate of appreciation of domestic currency, $(E_{t+1}^e - E_t)/E_t$.

3. As long as interest rates or the expected rate of appreciation are not too large, then, a good approximation implies:

$$\underset{2\%}{i_t} = \underset{5\%}{i_t^*} - \frac{E_{t+1}^e - E_t}{E_t} \tag{6}$$

$i = 2\%$

$i^* = 5\%$

4. This means that the domestic interest rate must be equal to foreign interest rate minus the expected rate of appreciation of the domestic currency.

5. Or equivalently, the domestic interest rate must be equal the foreign interest rate minus the expected rate of depreciation of the foreign currency.

$$\frac{1}{1+\frac{E_{t+1}^e - E_t}{E_t}}$$

$$= \frac{1}{\frac{\cancel{E_t} + E_{t+1}^e - \cancel{E_t}}{E_t}}$$

$$= \frac{E_t}{E_{t+1}^e}$$

The Balance of Payments

1. A country's international transactions are recorded in the **balance-of-payments**, BP.

2. BP has three main components:

   (a) Current account, CA

   (b) Financial account, FA, and

   (c) Capital account, KA.

3. Therefore, BP is usually written as:

$$BP = CA + FA + KA \tag{7}$$

4. Since capital account balance is usually negligible, we are not discussing it here. Hence,

$$BP = CA + FA \tag{8}$$

5. Any transaction resulting in a payment to foreigners is entered in the BoP accounts as a debit and is given a negative $(-)$ sign. Any transaction resulting in a receipt from foreigners is entered as a credit and is given a positive $(+)$ sign.

6. The current account records exports and imports of goods and services and international receipts or payments of income.

7. Exports and income receipts enter as positive. Imports and income payments enter as negative.

8. The financial account keeps records of sales of assets to foreigners (financial inflows, FI) and purchases of assets located abroad by domestic residents (financial outflows, FO).

9. Sales of assets to foreigners enter as positive. Purchases of assets located abroad enter as negative.

[Table 2: The US balance of payments in 2018 in billions of USD]

10. BoP follows a fundamental accounting principle known as *double-entry bookkeeping* with each transactions enters the BoP twice, once with a positive sign and another with a negative sign.

11. Because any international transactions automatically gives rise to two offsetting entries in the BoP, we have

$$\text{Current account balance} = -\text{ Financial account balance} \tag{9}$$

12. The current account, on the other hand, consists of three components:

$$\text{Current account} = \text{Trade balance} + \text{Income balance} + \text{Net unilateral transfer} \quad (10)$$

13. The trade balance or balance on goods and services is the difference between exports and imports of goods and services. It consists of

    (a) Merchandise trade balance or balance on goods, and

    (b) Services balance.

14. The merchansise trade balance keeps record of net exports of goods.

15. The services balance keeps record of net receipts from items such as transportation, travel expenditures and legal assistance.

16. The income balance has

    (a) Net investment income, and

    (b) Net investment compensation to employees.

17. Net investment income is the difference between income receipts on US-owned assets abroad and income payments on foreign-owned assets in the US. It includes items such as international interest, dividend payments and earnings of domestically owned firms operating abroad.

18. Net investment compensation to employees includes

    (a) As positive entries: (1) earnings of US residents employed temporarily abroad, (2) earnings of US residents employed by foreign governments in US, and (3) earnings of US residents employed by international organizations in US.

    (b) As negative entries: (1) foreign workers who commute to work in US, (2) foreign students studying in US, (3) foreign professionals temporarily residing in US, and (4) foreign temporary workers in US.

19. Net unilateral transfers keeps record of the difference between gifts. One common item is private remittances.

20. The financial account has two main components as follows:

$$\begin{aligned} \text{Financia account} = \; & \text{Increased in foreign-owned assets in the US} && (11) \\ & -\text{Increased in US-owned assets abroad} && (12) \end{aligned}$$

21. Note that foreign-owned assets in the US includes

(a) US securities held by foreign residents

(b) US currency held by foreign residents

(c) US borrowing from foreign banks

(d) Foreign direct investment in the US

22. US-owned assets abroad are

    (a) Foreign securities

    (b) US bank lending to foreigners

    (c) US direct investment abroad

23. Note that when US borrows $1 from foreigners, it is selling them an asset - a promise that tey will be repaid $1, with interest, in the future. Such transactions enters the financial account with a positive sign because the loan is itself a payment to the US or a financial inflow (also sometimes called a capital inflow).

24. When US lends abroad, however, a payment is made to foreigners. This transaction involves the purchase of an asset from foreigners and is called a financial outflow (or, alternatively, a capital outflow).

25. There is another type of transaction that merits separate discussion. This type of transaction is the purchase or sale of official reserve assets by central banks.

26. Official international reserves are foreign assets held by central banks as a cushion against national economic misfortune. At one time, official reserves consisted largely gold, but today central bank's reserves include substantial foreign financial assets.

27. Central banks often buy or sell international reserves in private asset markets to affect macroeconomic conditions in their economies. Official transactions of this type are called **official foreign exchange interventions.**

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Quiz date : 6 Sept 2024
430pm
Duration : ~~60~~ 75 mins
20marks 40 MCQs
~~10~~ 20 mins break
20 mins
5 marks Open ended Questions
2 Questions
Coverage : Lec 1 & Lec 2
only content before the Balance of Payments

LECTURE 2
The Goods Market in an Open Economy

READINGS

1. Olivier Blanchard *Macroeconomics,* 8th Edition (2021), Prentice Hall, Chapter 18.

INTRODUCTION

1. In this lecture we ask what the effects of fiscal and monetary policies on economic activity are, when both goods and financial markets are open to the rest of the world.

2. Specifically, we will

   ① $G\uparrow$ ② $Y^*\uparrow$ Y

   (a) Show the effects of domestic shocks and foreign shocks on the domestic economy's output and trade balance. TB

   ③ $\varepsilon\downarrow$ Y TB

   (b) Looks at the effects of a real depreciation on output and the trade balance.

THE IS RELATION IN THE OPEN ECONOMY

The Demand for Domestic Goods

1. In an open economy, the demand for domestic goods, Z, is given by:

$$ZZ \qquad Z = C + I + G + (X - \frac{IM}{\epsilon}) \qquad TB = NX = X - \frac{IM}{\varepsilon} \quad (1)$$

open

2. The first three terms – consumption, C, investment, I, and government spending, G – constitute the domestic demand for goods. If the economy were closed, $C + I + G$ would also be the demand for domestic goods.

3. Now we have to make two adjustments:

   (a) We must subtract imports – that part of the domestic demand that falls on foreign goods rather than on domestic goods. It should be noted that foreign goods are different from domestic goods, so we cannot just subtract the quantity of imports, IM. We must first express the value of imports in terms of domestic goods. This is what IM/ϵ in equation stands for. IM/ϵ is the value of imports in terms of domestic goods. $IM \cdot \frac{1}{\varepsilon}$

(b) We must add exports – that part of the demand for domestic goods that comes from abroad. This is captured by the term X in equation.

X

Note : i – nominal r – real

The Determinants of C, I and G

1. Domestic demand is given by:

DD

Y-T disposable income $(Y-T)\uparrow \rightarrow C\uparrow$

constant

$$C + I + G = C(\overset{+}{Y - T}) + I(\overset{+}{Y}, \overset{-}{r}) + G \tag{2}$$

$Y\uparrow \rightarrow I\uparrow$

r – cost of borrowing

2. It is noted that C depends positive on $(Y - T)$, and I depends positively on Y, and negatively on r.

$r\uparrow \rightarrow$ cost of borrowing$\uparrow$ $\rightarrow I\downarrow$

The Determinants of Imports IM

1. Imports clearly depend on domestic income: higher domestic income leads to a higher domestic demand for all goods, both domestic and foreign. So, a higher domestic income leads to higher imports.

2. Imports also clearly depend on the real exchange rate – the price of domestic goods in terms of foreign goods. The more expensive domestic goods are relative to foreign goods – equivalently, the cheaper foreign goods are relative to domestic goods – the higher the domestic demand for foreign goods. So a higher real exchange rate leads to higher imports. Thus, we write imports as: $\varepsilon\uparrow$ ① $E\uparrow$ ② $P\uparrow$ ③ $P^*\downarrow$

Note : $\varepsilon = \frac{EP}{P^*}$

$$IM = IM(\overset{+}{Y}, \overset{+}{\epsilon}) \tag{3}$$

① $Y\uparrow \rightarrow IM\uparrow$

② $\varepsilon\uparrow \rightarrow$ real appⁿ $\rightarrow$ X exp

$\rightarrow$ IM cheaper $\rightarrow IM\uparrow$

3. It is noted that:

   (a) An increase in domestic income, Y (equivalently, an increase in domestic output – income and output are still equal in an open economy), leads to an increase in imports.

   (b) An increase in the real exchange rate, ϵ i.e. an appreciation of real exchange rate, leads to an increase in imports, IM.

The Determinants of Exports

1. Exports are the part of foreign demand that falls on domestic goods.

$Y^*\uparrow \rightarrow C^*\uparrow$ & $X\uparrow$

2. They depend on foreign income: higher foreign income means higher foreign demand for all goods, both foreign and domestic. So higher foreign income leads to higher exports.

Y^*

3. Exports also depend on the real exchange rate: the higher the price of domestic goods in terms of foreign goods, the lower the foreign demand for domestic goods. In other words, the higher the real exchange rate, the lower the exports.

ε

4. We therefore write exports as:

$$X = X(\overset{+}{Y^*}, \overset{-}{\epsilon}) \tag{4}$$

where Y^* is foreign income.

(a) An increase in foreign income, Y^*, leads to an increase in exports.

(b) An increase in the real exchange rate, ϵ, leads to a decrease in exports.

$\varepsilon \uparrow \rightarrow$ real appⁿ $\rightarrow X$ expensive $\rightarrow X \downarrow$

Putting the Components Together

Slope of DD = MPC < 1

$Y\uparrow \begin{cases} \rightarrow C\uparrow \\ \rightarrow I\uparrow \end{cases} \rightarrow DD\uparrow$

DD upward.

1. The line DD plots domestic demand, $C + I + G$, as a function of output, Y.

2. The slope of the relation between demand and output is positive but less than 1. slope off of DD < 1

3. To arrive at the demand for domestic goods, we must first subtract imports. It gives us the line AA.

4. The line AA represents the domestic demand for domestic goods. The distance between DD and AA equals the value of imports, IM/ϵ.

5. Because the quantity of imports increases with income, the distance between the two lines increases with income.

6. We can establish two facts about line AA:

   (a) AA is flatter than DD: as income increases, some of the additional domestic demand falls on foreign goods rather than on domestic goods. In other words, as income increases, the domestic demand for domestic goods increases less than total domestic demand.

   (b) As long as some of the additional demand falls on domestic goods, AA has a positive slope: an increase in income leads to some increase in the demand for domestic goods.

7. Finally, we must add exports. This is done and it gives us the line ZZ, which is above AA.

8. The line ZZ represents the demand for domestic goods.

9. The distance between ZZ and AA equals exports. Because exports do not depend on domestic income (they depend on foreign income), the distance between ZZ and AA is constant, which is why the two lines are parallel.

10. Because AA is flatter than DD, ZZ is also flatter than DD.

11. This relation between net exports and output is represented as the line NX.

12. Net exports are a decreasing function of output: as output increases, imports increase, and exports are unaffected, so net exports decrease.

13. Call Y_{TB} the level of output at which the value of imports equals the value of exports, so that net exports are equal to 0.

14. Levels of output above Y_{TB} lead to higher imports and to a trade deficit. Levels of output below Y_{TB} lead to lower imports and to a trade surplus.

EQUILIBRIUM OUTPUT AND TRADE BALANCE

1. The goods market is in equilibrium when domestic output equals the demand – both domestic and foreign – for domestic goods:

AE in an open economy

$$Y = Z \tag{5}$$

2. Given the components of the demand for domestic goods, we get:

$$Y = C(Y-T) + I(Y,r) + G + \underbrace{X(Y^*,\epsilon) - \frac{IM(Y,\epsilon)}{\epsilon}}_{NX} \tag{6}$$

For the goods market to be in equilibrium, output (the left side of the equation) must be equal to the demand for domestic goods (the right side of the equation).

3. It will be convenient in what follows to regroup the last two terms under **net exports** defined as exports minus the value of imports:

$$NX(Y,Y^*,\epsilon) = X(Y^*,\epsilon) - \frac{IM(Y,\epsilon)}{\epsilon} \tag{7}$$

4. Using this definition of net exports, we can rewrite the equilibrium condition as:

$$Y = C(Y-T) + I(Y,r) + G + NX(Y,Y^*,\epsilon) \tag{8}$$

5. The main implication of the equation is that both the interest rate and the real exchange rate affect demand and, in turn, equilibrium output:

$r\uparrow \rightarrow I\downarrow \rightarrow ZZ\downarrow \rightarrow Y=Z,\ Y\downarrow$

(a) An increase in the real interest rate leads to a decrease in investment spending and, as a result, to a decrease in the demand for domestic goods. This leads, through the multiplier, to a decrease in output. $\epsilon\uparrow \rightarrow$ real appn $\rightarrow$ X exp $\rightarrow X\downarrow$; IM cheaper $\rightarrow IM\uparrow$ $\rightarrow NX = X - M \rightarrow NX\downarrow$

(b) An increase in the exchange rate leads to a shift in demand toward foreign goods and, as a result, to a decrease in net exports. The decrease in net exports decreases the demand for domestic goods. This leads, through the multiplier, to a decrease in output. $\rightarrow ZZ\downarrow \rightarrow Y=Z,\ Y\downarrow$

(c) As we are still studying the short run, when prices (P, P^*) are assumed to be constant, the real exchange rate, $\epsilon = EP/P^*$, and the nominal exchange rate, E, move together. A decrease in the nominal exchange rate – a nominal depreciation – leads, one-for-one, to a decrease in the real exchange rate – a real depreciation. Conversely, an increase in the nominal exchange rate – a nominal appreciation – leads, one-for-one, to an increase in the real exchange rate – a real appreciation. If, for notational convenience, we choose P and P^* so that $P/P^* = 1$, then $\epsilon = E$, and:

$\epsilon = E$

$\epsilon\uparrow \rightarrow E\uparrow$

$Y\downarrow$

Note: $i = r + \pi^e$, $\pi^e = 0$

$\epsilon = \frac{EP}{P^*}$, $\left(\frac{P}{P^*}\right) = 1$, $\epsilon = E$

$$Y = C(Y-T) + I(Y,r) + G + NX(Y,Y^*,E) \tag{9}$$

6. Goods market equilibrium implies that output depends negatively on both the nominal interest rate and the nominal exchange rate.

INCREASES IN DEMAND - DOMESTIC OR FOREIGN

Increases in Domestic Demand

1. Suppose the economy is in a recession, and the government decides to increase government spending in order to increase domestic demand and output. What will be the effects on output and on the trade balance?
2. Before the increase in government spending, demand is given by ZZ_0 and the equilibrium is at point 0, where output equals Y_0. Let's assume that trade is initially balanced.
3. What happens if the government increases spending by ΔG? At any level of output, demand is higher by ΔG, shifting the demand relation up by ΔG, from ZZ_0 to ZZ_1. The equilibrium point moves from 0 to 1, and output increases from Y_0 to Y_1.
4. The increase in output is larger than the increase in government spending: there is a **multiplier effect**.
5. Compared to a closed economy,
   (a) The increase in output from Y_0 to Y_1 leads to a trade deficit as imports increase and exports do not change.
   (b) Not only does government spending now generate a trade deficit, but the effect of government spending on output is smaller than it would be in a closed economy. This is because the multiplier is smaller in an open economy.

Increases in Foreign Demand $Y^* \uparrow$

1. Consider now an increase in foreign output, that is, an increase in Y^*. We want to analyse its effects on the domestic economy.

2. The initial demand for domestic goods is given by ZZ_0 and the equilibrium is at point 0, with output level Y_0.

3. Let's again assume that trade is balanced.

4. It will be useful here to refer to the line that shows the domestic demand for goods, $C + I + G$, as a function of income. This line is drawn as DD_0 which is steeper than ZZ_0.

5. The difference between ZZ_0 and DD_0 equals net exports, so that if trade is balanced at point 0, ZZ_0 and DD_0 intersect at point 0.

6. Now consider the effects of an increase in foreign output, ΔY^*.

7. Higher foreign output means higher foreign demand, including foreign demand for domestic goods.

8. So, the direct effect of the increase in foreign output is an increase in domestic exports by some amount, which we shall denote by ΔX.

   (a) For a given level of output, this increase in exports leads to an increase in the demand for goods by ΔX, so ZZ shifts up by ΔX, from ZZ_0 to ZZ_1.

   (b) For a given level of output, net exports go up by ΔX. So the line showing net exports also shifts up by ΔX, from NX_0 to NX_1

9. The new equilibrium is at point 1 with output level Y_1.

10. The increase in foreign output leads to an increase in domestic output.

11. Higher foreign output leads to higher exports of domestic goods, which increases domestic output and the domestic demand for goods through the multiplier.

12. What happens to the trade balance?

    (a) We know that exports go up. But could it be that the increase in domestic output leads to such a large increase in imports that the trade balance actually deteriorates?

    (b) No. The trade balance must improve.

    (c) Note that when foreign demand increases, the demand for domestic goods shifts up from ZZ_0 to ZZ_1; but the line DD_0, which gives the domestic demand for goods as a function of output, does not shift.

(d) At the new equilibrium level of output, Y_1, domestic demand is given by the distance DC, and the demand for domestic goods is given by $D1$. Net exports are therefore given by the distance $C1$ – which, because DD_0 is necessarily below ZZ_1, is necessarily positive.

(e) Thus, the increase in imports does not offset the increase in exports, and the trade balance improves.

DEPRECIATION, THE TRADE BALANCE AND OUTPUT

1. Suppose the US government takes policy measures that lead to a depreciation of the dollar – a decrease in the nominal exchange rate.

2. Recall that the real exchange rate is given by:

$$\epsilon = \frac{EP}{P^*} \tag{10}$$

3. The real exchange rate, ϵ (the price of domestic goods in terms of foreign goods), is equal to the nominal exchange rate, E (the price of domestic currency in terms of foreign currency), times the domestic price level, P, divided by the foreign price level, P^*.

4. In the short run, we can take the two price levels, P and P^*, as given. This implies that the nominal depreciation is reflected one-for-one in a real depreciation. $\epsilon = E$

5. How will this real depreciation affect the US trade balance and US output?

Depreciation and the Trade Balance: The Marshall-Lerner Condition

1. Recall that

$$NX = X - \frac{IM}{\epsilon} \tag{11}$$

$\varepsilon\downarrow$ → X cheaper → $(X)\uparrow$; → M expensive → $M\downarrow$

Replace X and IM accordingly so that:

$$NX = X(Y^*, \epsilon) - \frac{IM(Y, \epsilon)}{\epsilon} \tag{12}$$

2. This suggests that real ($\varepsilon\downarrow$) depreciation affects the trade (TB) balance through three channels:

   (a) Exports, X, increase. ($\varepsilon\downarrow \rightarrow X\uparrow$) The real depreciation makes domestic goods relatively less expensive abroad. This leads to an increase in foreign demand for domestic goods i.e an increase in domestic exports. $(X - \frac{IM}{\varepsilon})\uparrow$

   (b) Imports, IM, decrease. ($\varepsilon\downarrow \rightarrow M\downarrow$) The real depreciation makes foreign goods relatively more expensive in the domestic economy. This leads to a shift in domestic demand toward domestic goods and to a decrease in the quantity of imports. $\varepsilon\downarrow \rightarrow \frac{1}{\varepsilon}\uparrow \rightarrow \frac{IM}{\varepsilon}\uparrow$

   SR (c) The relative price of foreign goods in terms of domestic goods, $1/\epsilon$, increases. This increases the import bill, IM/ϵ. The same quantity of imports now costs more to buy.

3. For the trade balance to improve following a depreciation, exports must increase enough and imports must decrease enough to compensate for the increase in the price of imports. The condition under which a real depreciation leads to an increase in net exports is known as the **Marshall–Lerner condition**. → $\varepsilon\uparrow \rightarrow NX\uparrow$

The Effects of a Real Depreciation

1. We have looked so far at the direct effects of a depreciation on the trade balance.

2. But the effects do not end there.

   $NX\uparrow$ $Y\uparrow$

3. The change in net exports changes domestic output, which affects net exports further.

4. Because the effects of a real depreciation are very much like those of an increase in foreign output, a depreciation leads to an increase in net exports (assuming that the Marshall–Lerner condition holds), at any level of output. Recall $Y^* \uparrow \rightarrow X\uparrow \rightarrow NX\uparrow \rightarrow ZZ\uparrow \rightarrow \cdots$ case 2.

   $\varepsilon\downarrow \rightarrow X\uparrow \,\&\, M\downarrow \rightarrow NX\uparrow \rightarrow ZZ\uparrow$ case 3

5. Both the demand relation (ZZ) and the net exports relation (NX) shift up.

6. The equilibrium moves from point 0 to point 1, and output increases from Y_0 to Y_1.

7. The trade balance improves: the increase in imports induced by the increase in output is smaller than the direct improvement in the trade balance induced by the depreciation.

8. In summary, the depreciation leads to a shift in demand, both foreign and domestic, toward domestic goods. This shift in demand leads in turn to both an increase in domestic output and an improvement in the trade balance.

9. Although a depreciation and an increase in foreign output have the same effect on domestic output and the trade balance, there is a subtle but important difference between the two.

10. A depreciation works by making foreign goods relatively more expensive. But this means that given their income, people who now have to pay more to buy foreign goods because of the depreciation, are worse off.

11. This mechanism is strongly felt in countries that go through a large depreciation. Governments trying to achieve a large depreciation often find themselves with strikes and riots in the streets, as people react to the much higher prices of imported goods.

$Y = Y_f$ and $TB = 0$ or $NX = 0$

Combining Exchange Rate and Fiscal Policies $Y = Y_f$ ✓ at $Y = Y_f$, $TB < 0$ ✗

1. Suppose output is at its natural level, but the economy is running a large trade deficit. The government would like to reduce the trade deficit while leaving output unchanged. What should it do?

2. A depreciation alone will not do: it will reduce the trade deficit, but it will also increase output.

3. Nor will a fiscal contraction work: it will reduce the trade deficit, but it will decrease output.

4. What should the government do?

5. The answer: use the right combination of depreciation and fiscal contraction.

6. Suppose the initial equilibrium in is at point 0, associated with output Y_0. At this level of output, there is a trade deficit, given by the distance BC. If the government wants to eliminate the trade deficit without changing output, it must do two things:

   (a) It must achieve a depreciation sufficient to eliminate the trade deficit at the initial level of output. The depreciation must be such as to shift the net exports relation from NX_0 to NX_1. The problem is that this depreciation, and the associated increase in net exports, also shifts the demand relation from ZZ_0 to ZZ_1. In the absence of other measures, the equilibrium would move from point 0 to point 1, and output would increase from Y_0 to Y_1.

   (b) In order to avoid the increase in output, the government must reduce government spending so as to shift ZZ_1 back to ZZ_0. This combination of a depreciation and a fiscal contraction leads to the same level of output and an improved trade balance.

7. There is a general point behind this example: to the extent that governments care about both the level of output and the trade balance, they have to use both fiscal policy and exchange rate policies.

SAVING, INVESTMENT, AND THE CURRENT ACCOUNT BALANCE

1. Recall that the equilibrium in the goods market suggests that investment equals to saving, which is the sum of private saving and public saving.

2. We can now derive the corresponding condition for the open economy.

3. At equilibrium,

$$Y = C + I + G + (X - \frac{IM}{\epsilon}) \tag{13}$$

4. Rearrange gives:

$$Y - T - C = I + (G - T) + NX \tag{14}$$

   where $NX = X - IM/\epsilon$.

5. Recall that, in an open economy, the total income of domestic residents is equal to domestic income, which is equal to domestic output, Y, plus net income from abroad, NI.

6. Add NI to both sides of the equation gives:

$$(Y + NI - T) - C = I + (G - T) + (NX + NI) \tag{15}$$

   where $Y + NI - T$ is the private saving, S_p and $T - G$ is the public saving, S_g and $NX + NI$ is the current account, CA.

7. As a result,

$$S_p = I + (G - T) + CA \tag{16}$$

8. Or,

$$CA = S_p + (T - G) - I \tag{17}$$

9. Given national saving, $S = S_p + S_g$, then:

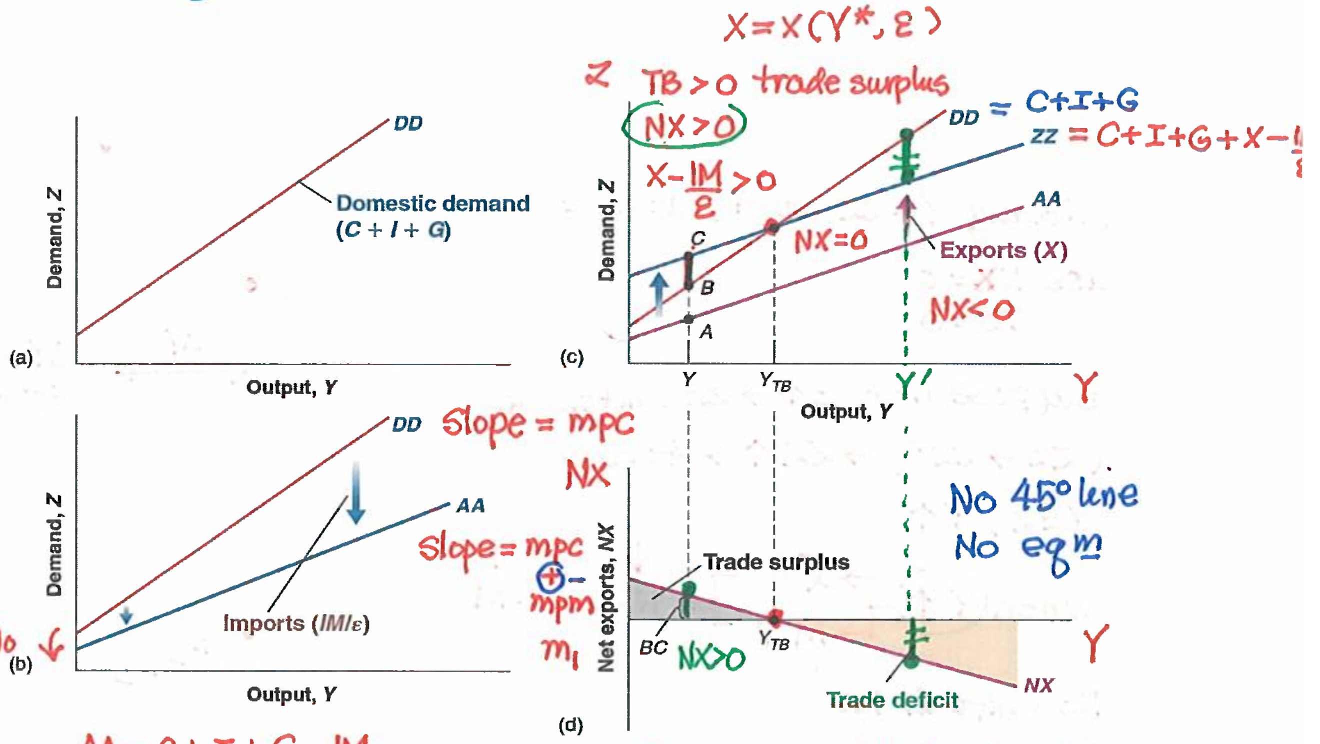
$$CA = S - I \tag{18}$$

10. The current account balance is equal to national saving (private plus public) minus investment.

    (a) A current account **surplus** implies that the country is saving **more than** it invests.

    (b) A current account **deficit** implies that the country is saving **less than** it invests.

18.1 The *IS* Relation in the Open Economy



$AA = C + I + G - \frac{IM}{\varepsilon}$

$IM = IM(\overset{+}{Y}, \varepsilon) = m_0 + m_1 Y$

MPM

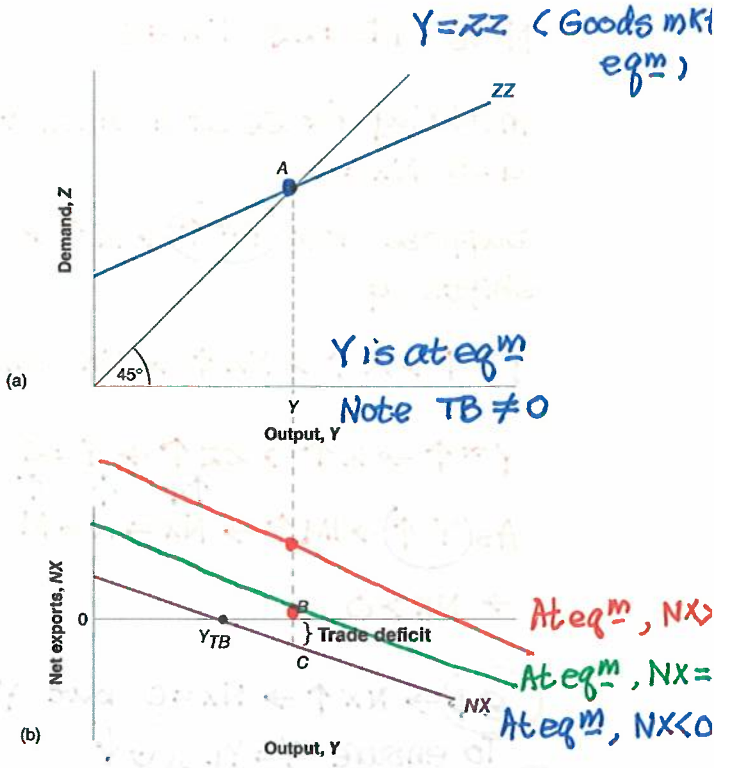
Recall $X = X(Y^*, \varepsilon)$ $IM = IM(Y, \varepsilon)$

NX downward ↓

$Y\uparrow \rightarrow IM\uparrow \rightarrow NX = \bar{X} - \overset{\uparrow}{IM} \rightarrow NX\downarrow$

Pearson

18.2 Equilibrium Output and the Trade Balance



$i = r + \overset{0}{\pi^e}$

Pearson

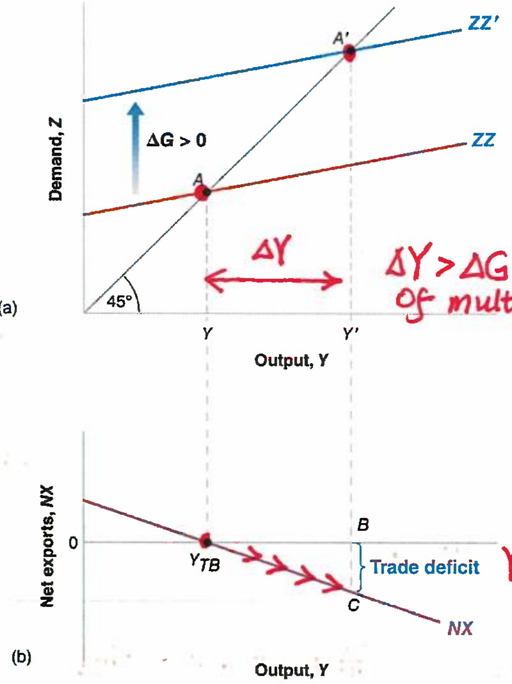
18.3 Increases in Demand–Domestic or Foreign

Initially Y=ZZ at A, at eqm
ass NX=0
TB=0

Suppose now G↑ → ZZ shifts up (ZZ → ZZ') → ZZ↑ → At eqm Y=Z, Y↑

when Y↑ → X̄ (↓), M↑ (↑) → NX = X̄ − M

No shift in NX



ΔY

ΔY > ΔG b/c of multiplier

Y

18.3 Increases in Demand–Domestic or Foreign

Note: DD = ZZ NX = 0

Initially Y = ZZ at A. At eqm
ass NX = 0

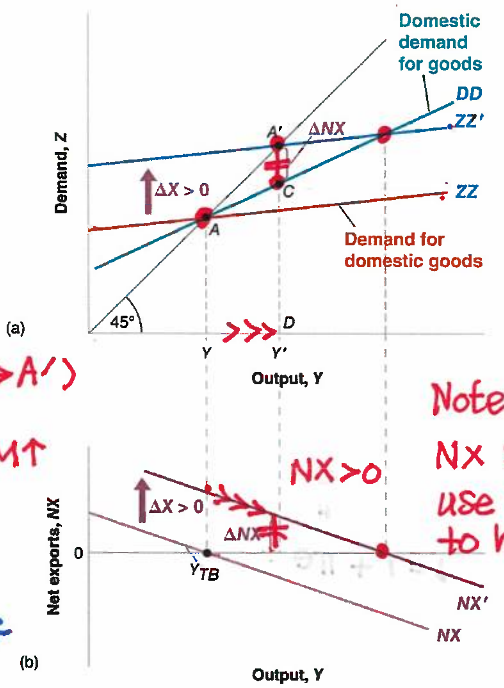
Suppose now Y*↑ → X↑ → ZZ shifts up.

Y*↑ → X↑ → NX↑ → NX shifts up

Y*↑ → X↑ → ZZ↑ → Y = Z, Y↑ (A → A')

As Y↑ → IM↑ → NX = X − M → X↑ > M↑
→ NX > 0

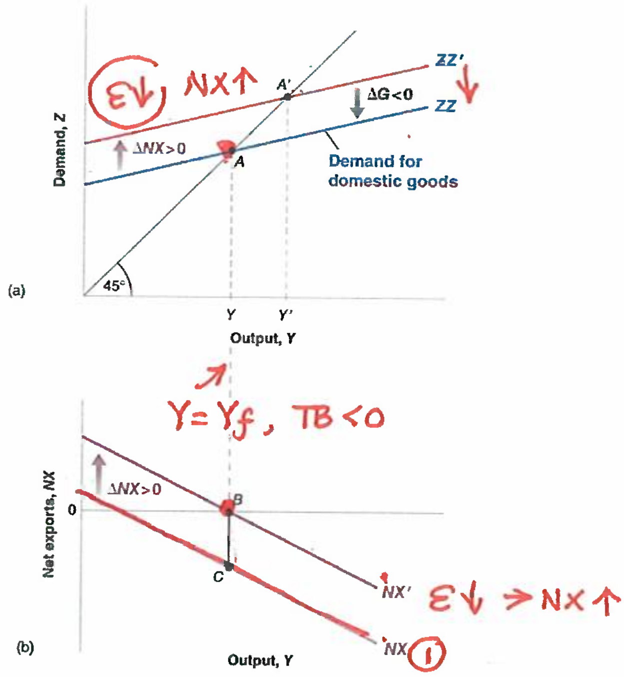
ε↓ → NX↑ → NX = 0 but Y↑ → Y > Y_f
To ensure Y = Y_f, G↓



NX > 0

Note: NX shifts use DD line to help

18.4 Depreciation, the Trade Balance, and Output



Pearson

APPENDIX: Derivation of the Marshall-Lerner Condition

- The Marshall-Lerner condition is the condition under which a real depreciation (lower *U*) leads to an increase in net exports.
- Given the definition of net exports: $NX = X - IM/\varepsilon$
- Multiply both sides by ε to get: $\varepsilon NX = \varepsilon X - IM$, which can be written as: $\varepsilon(\Delta NX) = (\Delta\varepsilon)X + \varepsilon(\Delta X) - (\Delta IM)$
- Divide both sides by εX to get:

$$\frac{[\varepsilon(\Delta NX)]}{\varepsilon X} = \frac{[(\Delta\varepsilon)X]}{\varepsilon X} + \frac{[\varepsilon(\Delta X)]}{\varepsilon X} - \frac{[\Delta IM]}{\varepsilon X}$$

- If trade is initially balanced ($\varepsilon X = IM$), then:

$$\frac{(\Delta NX)}{X} = \frac{(\Delta\varepsilon)}{\varepsilon} + \frac{(\Delta X)}{X} - \frac{\Delta IM}{IM}$$

Pearson